

“I envision a time when we’ll be to robots what dogs are to humans — loyal, slightly confused.”
Claude Shannon.

IAG–USP Postgraduate Mini-Course

Machine Learning, Deep Learning, and AI for Astrophysics

Exercise Set 1

Foundations, Deep Learning, and Scientific Reliability

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1 General instructions

This problem set is designed to reinforce the conceptual framework of the first week of the mini-course. Unless a question explicitly asks for numerical computation, concise but technically precise answers are preferred. Students should justify methodological choices in astrophysical terms rather than merely naming an algorithm.

Learning goals

By the end of this set, students should be able to: formulate astrophysical questions as statistical-learning problems; reason about losses, metrics, validation, and class imbalance; explain the main inductive biases of common neural architectures; identify leakage, shortcut learning, and domain shift; reason about calibration and uncertainty; and design an end-to-end, scientifically defensible ML workflow.

Notation. A supervised dataset is written as

$$\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N ,$$

where $\mathbf{x}_i$ denotes the measured representation of object i and y_i the target. A parametric predictor is denoted f_θ , with empirical risk

$$\hat{R}(\theta) = \frac{1}{N} \sum_{i=1}^N \mathcal{L}(f_\theta(\mathbf{x}_i), y_i) .$$

2 Monday: Statistical learning for astrophysical inference

2.1 Slot 1: From scientific questions to learning problems

Exercise 1 — Identify the learning paradigm

For each scientific question below, identify the most natural primary formulation among *classification*, *regression*, *clustering*, *anomaly detection*, *representation learning*, *density estimation*, *simulation-based inference*, or *surrogate modeling*. Briefly justify your answer.

- Decide whether a transient is a Type Ia supernova, core-collapse supernova, tidal disruption event, or another class.
- Estimate stellar effective temperature from a low-resolution spectrum.
- Search an all-sky catalog for objects whose broadband spectral energy distributions are unlike the bulk population.
- Compress millions of unlabeled stellar spectra into a low-dimensional latent representation before any downstream label is defined.
- Infer a posterior distribution for cosmological parameters when the likelihood is unavailable but forward simulations are possible.
- Replace an expensive radiative-transfer calculation with a fast emulator.
- Discover previously unknown phenomenological groups among variable-star light curves.
- Estimate the probability that a B-type star hosts a circumstellar disk.

Exercise 2 — Prediction is not explanation

A model predicts stellar age from spectra with excellent test-set accuracy.

- Does this imply that the features used by the model are causally responsible for stellar aging? Explain.
- Give two examples of nuisance variables or confounders that could support strong prediction without representing the physical mechanism of interest.
- Propose one diagnostic that could help determine whether the model is exploiting an astrophysically meaningful signal or a survey-dependent shortcut.

Exercise 3 — Construct the data model

Suppose each object has an irregular light curve $\{(t_j, f_j, \sigma_j)\}_{j=1}^{n_i}$, sky coordinates, a survey identifier, and a binary scientific label.

- Write a data model that distinguishes the scientific target, observational data, metadata, and provenance variables.
- Which variables should be considered candidate inputs to the model, and which should initially be used only for auditing?
- Explain why feeding the survey identifier directly to a classifier may improve validation accuracy while harming scientific validity.

Challenge 1 — What is the actual target?

In many astronomical problems, the label is not the physical state itself but a catalog decision made from previous measurements. Discuss the distinction between learning $P(y_{\text{catalog}} | x)$ and learning a physical probability $P(z_{\text{physical}} | x)$. What assumptions are required to interpret the former as evidence about the latter?

2.2 Slot 2: Losses, validation, overfitting, and metrics**Exercise 4 — Empirical risk and weighted binary cross-entropy**

For binary classification, let the model output $p_i = P_\theta(y_i = 1 | \mathbf{x}_i)$.

- Write the ordinary binary cross-entropy loss.
- Write a weighted version with positive-class weight α and negative-class weight β .
- Explain how increasing α/β changes the optimization pressure.
- Why is class weighting not, by itself, a solution to dataset shift?

Exercise 5 — Rare-object metrics

A test set contains 10,000 objects, of which 200 are genuine members of a rare astrophysical class. A classifier produces

$$\text{TP} = 150, \quad \text{FN} = 50, \quad \text{FP} = 300, \quad \text{TN} = 9500.$$

Compute:

- accuracy;
- precision;
- recall;
- F1 score;
- specificity;
- balanced accuracy.

Then answer: would you call this model “good” if the scientific goal is to construct an expensive spectroscopic follow-up list? What if the goal is to miss as few genuine rare objects as possible?

Exercise 6 — Why accuracy can be misleading

Consider a survey with 98,000 ordinary objects and 2,000 rare objects.

- What accuracy is obtained by predicting the majority class for every object?
- What are the rare-class recall and F1 score of this trivial classifier?
- Explain why precision–recall curves are often more informative than ROC curves in strongly imbalanced candidate-search problems.

Exercise 7 — Data splitting and leakage

For each scenario, state whether the proposed split is scientifically acceptable and explain the main risk.

- Multiple observations of the same star are randomly distributed across train and test sets.
- A set of simulated spectra generated from the same physical parameter grid point appears in both train and test sets with different noise realizations.
- Hyperparameters are repeatedly changed after inspecting the final test-set score.
- Training uses Survey A and testing uses Survey B, with no objects in common.
- The model is trained on one observing season and tested on a later season of the same objects.

For each unacceptable case, propose a safer split strategy.

Exercise 8 — Bias, variance, and regularization

A neural network has very low training error but much worse validation error.

- Is this primarily a bias or variance problem?
- Give four interventions that could plausibly improve generalization.
- Which intervention would be most suspicious if it improves validation performance only because it removes physically difficult objects from the dataset?

Exercise 9 — Heteroscedastic regression

Suppose a model predicts a scalar astrophysical parameter y and also predicts an object-dependent uncertainty $\sigma_\theta(\mathbf{x}) > 0$. Assuming a Gaussian conditional model,

$$y \mid \mathbf{x} \sim \mathcal{N}(\mu_\theta(\mathbf{x}), \sigma_\theta^2(\mathbf{x})) ,$$

show that, up to an additive constant, the negative log-likelihood for one object can be written as

$$\mathcal{L} = \frac{(y - \mu_\theta)^2}{2\sigma_\theta^2} + \log \sigma_\theta .$$

Explain why the $\log \sigma_\theta$ term is necessary.

Exercise 10 — Model selection protocol

You have three candidate models and a limited labeled dataset.

- Design a train/validation/test workflow that allows hyperparameter tuning without contaminating the final result.
- When would nested cross-validation be justified?
- Why should preprocessing operations such as scaling, imputation, and feature selection be fitted only on the training partition?

3 Wednesday: Deep learning and representation learning

3.1 Slot 3: Neural networks, optimization, and learned representations

Exercise 11 — Parameter counting in a multilayer perceptron

Consider an MLP with input dimension 20, two hidden layers of widths 64 and 32, and a single sigmoid output.

- Compute the total number of trainable weights and biases.
- If the training set contains only 400 labeled objects, what concern does the parameter count raise?
- Explain why parameter count alone does not completely determine overfitting behavior.

Exercise 12 — One forward pass

Consider a single hidden layer

$$\mathbf{h} = \text{ReLU}(W\mathbf{x} + \mathbf{b}) , \quad \hat{y} = \sigma(\mathbf{v}^\top \mathbf{h} + c) .$$

Let

$$\mathbf{x} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} , \quad W = \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix} , \quad \mathbf{b} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} , \quad \mathbf{v} = \begin{pmatrix} 1 \\ -2 \end{pmatrix} , \quad c = 0 .$$

Compute $\mathbf{h}$ and $\hat{y}$.

Exercise 13 — Backpropagation as the chain rule

For a scalar model

$$h = \tanh(wx + b) , \quad \hat{y} = vh + c ,$$

with squared loss $L = \frac{1}{2}(\hat{y} - y)^2$, derive $\partial L / \partial w$. Identify explicitly the local derivatives multiplied by the chain rule.

Exercise 14 — Activation functions

Compare ReLU, sigmoid, and tanh in terms of saturation, gradient flow, output range, and likely use inside a modern deep network. Give one situation in which a sigmoid output remains appropriate.

Exercise 15 — Optimization diagnostics

You observe the following training behaviors. Suggest the most plausible causes and at least one intervention for each.

- Training and validation losses are both high and nearly constant.
- Training loss decreases rapidly, validation loss decreases initially and then rises.
- Loss becomes NaN after a few batches.
- Training loss oscillates strongly from epoch to epoch.
- Training is stable but extremely slow.

Exercise 16 — Representation learning

A traditional pipeline maps each light curve to 30 hand-designed features before classification. A deep sequence model instead learns a latent vector $\mathbf{h} = E_\theta(x)$.

- What does it mean to call $\mathbf{h}$ a learned representation?
- Give two potential advantages and two potential disadvantages relative to engineered features.
- Propose one diagnostic for testing whether $\mathbf{h}$ carries strong information about survey identity.

3.2 Slot 4: CNNs, sequence models, and transformers**Exercise 17 — Convolutional inductive bias**

Explain why a CNN is naturally suited to astronomical images. Your answer should explicitly discuss local connectivity, weight sharing, and translation equivariance.

- Is exact translation invariance always desirable in astronomy?
- Give one astrophysical imaging problem in which absolute position may carry meaningful information.

Exercise 18 — Convolution parameter count

A 1D convolution receives 4 input channels and produces 32 output channels with kernel width 5 and one bias per output channel.

- How many trainable parameters does the layer have?
- Compare this with a dense layer that maps a flattened sequence of length 500 with 4 channels to 32 outputs.
- What structural assumption allows the convolution to use far fewer parameters?

Exercise 19 — Receptive field

Consider three stacked 1D convolutional layers, each with kernel size 3, stride 1, and no dilation.

- What is the receptive-field width of one unit in the third layer?
- How would dilation help a temporal convolutional network capture long-timescale variability without an extremely deep stack?

Exercise 20 — Irregular light curves

You want to train a model on astronomical light curves with nonuniform cadence and varying sequence length. Compare the following representations:

- interpolation onto a regular grid;
- sequences of $(\Delta t_j, f_j, \sigma_j, \text{mask}_j)$;
- engineered summary features;
- a set-based or attention-based representation that directly receives irregular samples.

Discuss what information each may lose or distort.

Exercise 21 — RNN, TCN, or Transformer?

For each problem, choose a plausible starting architecture and justify the choice. More than one answer may be defensible.

- 256×256 galaxy images.
- 3,000-pixel normalized stellar spectra.
- Light curves with 5,000 regularly sampled points and long-range correlations.
- Sparse irregular transient light curves with uncertainty estimates and missing epochs.
- A multimodal problem combining spectra, catalog metadata, and time series.

Exercise 22 — Self-attention by hand

Let

$$Q = K = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad V = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix},$$

with $d_k = 2$. Compute

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{2}}\right),$$

row-wise and then compute AV . Interpret qualitatively what each output token is doing.

Exercise 23 — Positional information

Why does a vanilla self-attention layer not know the order of tokens? Discuss the role of positional encodings. For irregular time series, explain why encoding the actual observation times may be preferable to using only integer sequence positions.

Challenge 2 — Symmetry and representation choice

Choose one astronomical problem and identify a physical or observational symmetry that a model should ideally respect. Examples include translations, rotations, permutations of unordered measurements, periodic phase, or sign/angle conventions. Propose an architecture or input representation that incorporates this symmetry rather than forcing the network to learn it from data.

4 Friday: Scientific reliability, uncertainty, and modern AI

4.1 Slot 5: Real-data pathologies and trustworthy inference

Exercise 24 — Domain shift taxonomy

Distinguish *covariate shift*, *label shift*, and *concept shift*. For each, provide an astrophysical example involving two surveys or observing programs.

Exercise 25 — Source shortcut learning

Two light-curve datasets have very different class fractions and different cadences. A pooled classifier achieves excellent random-split performance.

- Explain why the result may be misleading.
- Design a source-prediction diagnostic.
- Propose three evaluation protocols that would provide stronger evidence of astrophysical generalization.
- If a model can predict survey identity with 99% accuracy from the same inputs, does this prove the class model is invalid? Explain what it does and does not establish.

Exercise 26 — Calibration

A classifier assigns probability 0.9 to 100 candidates, but only 60 are genuine positives.

- Is the model well calibrated in this probability region?
- What empirical positive fraction would you expect from a calibrated model among predictions near 0.9?
- Explain the difference between discrimination and calibration.
- Name two post-hoc calibration strategies.

Exercise 27 — Brier score

For four binary objects, the predicted positive-class probabilities and labels are

$$\mathbf{p} = (0.9, 0.7, 0.4, 0.1) , \quad \mathbf{y} = (1, 0, 1, 0) .$$

Compute the Brier score

$$\text{BS} = \frac{1}{N} \sum_i (p_i - y_i)^2 .$$

Which individual prediction contributes most to the score?

Exercise 28 — Aleatoric and epistemic uncertainty

Classify each source of uncertainty as primarily aleatoric, epistemic, or a mixture.

- a) Photon-counting noise in a faint spectrum.
- b) Prediction for a class that is poorly represented in the training sample.
- c) Unknown calibration systematics in one survey.
- d) Intrinsically stochastic variability in an accreting source.
- e) Extrapolation to a region of parameter space absent from training.

For each case, state whether collecting more data of the same kind is expected to reduce the uncertainty substantially.

Exercise 29 — Physically valid augmentation

For a stellar light-curve classification problem, assess whether each augmentation is likely safe, unsafe, or context-dependent. Justify physically.

- a) Mild Gaussian noise consistent with reported photometric uncertainty.
- b) Randomly permuting all timestamps while keeping flux values fixed.
- c) Cropping a long time series to shorter windows.
- d) Multiplying all fluxes by a constant factor.
- e) Time reversal.
- f) Random phase shift for a strictly periodic folded light curve.

Exercise 30 — Interpretability is not causality

Suppose SHAP ranks color index as the most important feature in a Be-star classifier.

- a) What can you legitimately conclude?
- b) What can you *not* conclude?
- c) Give one experiment that could test whether color is acting as a physically meaningful discriminator or merely reproducing a selection criterion used to construct the labels.

4.2 Slot 6: Transfer learning, self-supervision, and generative AI**Exercise 31 — Transfer learning**

A transformer has been pretrained on millions of unlabeled astronomical light curves, while your target problem has only 1,000 labeled examples.

- a) Compare linear probing, partial fine-tuning, and full fine-tuning.
- b) Which approach is least likely to overfit the small labeled set?
- c) What evidence would convince you that the pretrained representation transfers across surveys rather than memorizing survey-specific patterns?

Exercise 32 — Self-supervised objectives

Design one plausible self-supervised pretraining objective for each data type:

- a) stellar spectra;
- b) irregular light curves;
- c) galaxy images.

For each objective, state what kind of astrophysical representation you hope the encoder learns.

Exercise 33 — Diffusion-model intuition

In a standard forward diffusion process,

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I).$$

- a) What happens to the contribution of x_0 as $\bar{\alpha}_t \rightarrow 0$?
- b) What is the signal-to-noise ratio associated with this expression, up to a common variance convention?
- c) Why is learning the reverse process potentially useful for denoising, conditional generation, or simulation-based inference?
- d) Give one scientific risk of using generated astronomical data as if they were independent observations.

Exercise 34 — Foundation models and downstream validation

A foundation model produces a 512-dimensional embedding for every spectrum.

- a) Propose three downstream tasks that could use the embedding without retraining the encoder.
- b) Why is strong performance on an in-survey benchmark insufficient evidence that the embedding is “general astrophysical knowledge”?
- c) Design one cross-domain test that would provide stronger evidence.

5 Integrative exercises

Exercise 35 — Design a defensible Be-star classification study

You have two labeled light-curve datasets with different class fractions, different sampling properties, and partially different label semantics. A third table identifies color-selected B-star candidates but is not itself a Be/non-Be label source.

Design an end-to-end study. Your answer should include:

- a canonical object/label table;
- at least two explicit supervised tasks;
- train/validation/test splitting rules;
- one classical baseline and one deep model;
- a strategy for class imbalance;
- a diagnostic for source shortcut learning;
- appropriate primary metrics;
- a calibration strategy;
- an astrophysical error-analysis plan;
- one negative result that would still be scientifically informative.

Exercise 36 — Polarimetry as a representation problem

Linear polarization can be represented by normalized Stokes parameters

$$q = Q/I, \quad u = U/I,$$

or by

$$p = \sqrt{q^2 + u^2}, \quad \chi = \frac{1}{2} \text{atan2}(u, q).$$

- Why can the representation (q, u) be easier for a learning algorithm than (p, χ) ?
- Explain the angular periodicity of χ and why using $\sin 2\chi$ and $\cos 2\chi$ may be advantageous.
- Propose a regression task for inferring disk inclination from polarimetry.
- Propose a multimodal extension combining polarimetry and photometry.
- What uncertainty issue arises when computing p from noisy q and u , especially near low polarization?

Final challenge — Would you trust the model?

Choose one astrophysical ML result from a paper, your own research, or a hypothetical example. Write a one-page referee-style assessment addressing: data provenance; label definition; leakage risk; class balance; train/test distribution match; uncertainty; calibration; interpretability; baseline strength; physical plausibility; and whether the evidence is sufficient to support the claimed astrophysical conclusion.